Problem Set 6: MLE

1. Let $\ell = -\log(\sigma) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - x_i\beta)^2$. Find the value of μ that maximizes ℓ by setting $\ell'(\mu) = 0$ and solving. Check that the second derivative of $\ell(\mu)$ is negative at that value of μ; why do we care? Find the value of σ that maximizes ℓ by setting $\ell'(\sigma) = 0$ and solving.

2. Let $X_1, \ldots, X_n$ be i.i.d. $N(\mu, \sigma^2)$, and you have drawn a sample with observed values $(3.0, 5.0, 9.0)$.

   a. Write the log-likelihood $\ell(\mu, \sigma^2)$ for this sample. Then maximize it to find $\hat{\mu}$ and $\widehat{\sigma^2}$

   b. Compute a 95% confidence interval for μ

   c. Provide a 90% prediction interval for X.

3. Let y be an $n \times 1$ vector with $y \sim N(X\beta, \sigma^2 I)$, where X is $n \times p$: n observations of p predictors.

   a. Write the log-likelihood $\ell(\beta, \sigma^2)$ in matrix form.

   b. Show that maximizing ℓ over β is the same as minimizing $(y - X\beta)^\top (y - X\beta)$, and that $\hat{\beta} = (X^\top X)^{-1} X^\top y$.

   c. Take $n = 3$, $y = (2, 3, 5)^\top$, and

   $$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}.$$

   Compute $\hat{\beta}$. *Hint:* for a 2×2 matrix,

   $$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$